

Impaired fatty acid import or catabolism in macrophages restricts intracellular growth of *Mycobacterium tuberculosis*

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This **important** study reveals that disrupting fatty acid metabolism in macrophages significantly restricts the growth of *Mycobacterium tuberculosis*, showing that impaired lipid processing triggers various antimicrobial responses. Whilst the approach is robust, utilizing CRISPR-Cas9 knockout of multiple genes involved in lipid metabolism which yielded some **convincing** data, there are aspects that require improvement such as the autophagy assay and redox measurements. This work highlights how host lipid metabolism affects the ability of tubercle bacilli to thrive intracellularly, pointing to potential new therapeutic targets.

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Abstract

Mycobacterium tuberculosis (*Mtb*) infection of macrophages reprograms cellular metabolism to promote lipid retention. While it is clearly known that intracellular *Mtb* utilize host derived lipids to maintain infection, the role of macrophage lipid processing on the bacteria's ability to access the intracellular lipid pool remains undefined. We utilized a CRISPR-Cas9 genetic approach to assess the impact of sequential steps in fatty acid metabolism on the growth of intracellular *Mtb*. Our analyzes demonstrate that mutated macrophages that cannot either import, store or catabolize fatty acids restrict *Mtb* growth by both common and divergent anti-microbial mechanisms, including increased glycolysis, increased oxidative stress, production of pro-inflammatory cytokines, enhanced autophagy and nutrient limitation. We also show that impaired macrophage lipid droplet biogenesis is restrictive to *Mtb* replication, but increased induction fails to rescue *Mtb* growth. Our work expands our understanding of how host fatty acid homeostasis impacts *Mtb* growth in the macrophage.

Introduction

Mycobacterium tuberculosis (*Mtb*), the causative agent of tuberculosis (TB), has caused disease and death in humans for centuries (WHO, 2023). *Mtb* primarily infects macrophages in the lung (Cohen et al., 2018; Wolf et al., 2007) wherein the bacterium relies on host derived fatty acids and cholesterol for synthesis of its lipid rich cell wall, and to produce energy and virulence factors (Peyron et al., 2008; Russell et al., 2009; Singh et al., 2012; Daniel et al., 2011; Muñoz-Elías and McKinney, 2005; Pandey and Sasseti, 2008; Brzostek et al., 2009). Within the lung microenvironment, resident alveolar macrophages preferentially oxidize fatty acids and are more permissive to *Mtb* growth while recruited interstitial macrophages are more glycolytic and restrictive of *Mtb* replication (Huang et al., 2018; Pisu et al., 2020; Pisu et al., 2021). Globally, *Mtb* infection modifies macrophage metabolism in a manner that enhances its survival. *Mtb* infected macrophages shift their mitochondrial substrate preference to exogenous fatty acids which drives the formation of foamy macrophages that are laden with cytosolic lipid droplets (Peyron et al., 2008; Russell et al., 2009; Cumming et al., 2018; Singh et al., 2012; Podinovskaia et al., 2013). Foamy macrophages are found in abundance in the central and inner layers of granulomas, a common histopathological feature of human TB (Russell et al., 2009; Kim et al., 2010). Interference with key regulators of lipid homeostasis, such as the miR-33 and the transcription factors peroxisome proliferator-activated receptor α (PPAR α) and PPAR- γ enhances macrophage control of *Mtb* (Kim et al., 2017; Almeida et al., 2009; Ouimet et al., 2016). Moreover, compounds which modulate lipid metabolism such as the anti-diabetic drug metformin and some cholesterol lowering drugs are under investigation for host directed therapeutics (HDTs) against *Mtb* (Parihar et al., 2014; Singhal et al., 2014). Although the dependence of intracellular *Mtb* on host fatty acids and cholesterol is well documented (Wilburn et al., 2018), the impact of specific aspects of macrophage lipid metabolism on the bacteria remains opaque. In *Mtb* infected foamy macrophages, bacteria containing phagosomes are found in close apposition to intracellular lipid droplets (Peyron et al., 2008). It is believed that the bacterial induction of a foamy macrophage phenotype in host cells results in a steady supply of lipids that addresses the bacteria's nutritional requirements (Peyron et al., 2008; Russell et al., 2009; Singh et al., 2012; Daniel et al., 2011). In fact, intracellular *Mtb* has been shown to import fatty acids from host lipid droplet derived triacylglycerols (Daniel et al., 2011). However, other studies indicate that macrophage lipid droplet formation in response to *Mtb* infection can lead to the induction of a protective, anti-microbial response (Knight et al., 2018). *Mtb* appears unable to acquire host lipids when lipid droplets are induced by stimulation with interferon gamma (IFN- γ) (Knight et al., 2018). Moreover, there is some evidence that lipid droplets can be sites for the production of host protective pro-inflammatory eicosanoids (Knight et al., 2018; Daniel et al., 2011). Lipid droplets can also act as innate immune hubs against intracellular bacterial pathogens by clustering anti-bacterial proteins (Bosch et al., 2020). Inhibition of macrophage fatty acid oxidation by knocking out mitochondrial carnitine palmitoyl transferase 2 (CPT2) or using chemical inhibitors of CPT2 also restrict intracellular growth of *Mtb* (Chandra et al., 2020; Huang et al., 2018). These data demonstrate that modulation of the different stages in lipid metabolism inside *Mtb* infected macrophages can result in opposing outcomes.

We carried out a candidate-based, CRISPR mediated knockout of lipid import and metabolism genes in macrophages to determine their roles in intracellular growth of *Mtb*. By targeting genes involved in fatty acid import, sequestration, and metabolism in Hoxb8 derived conditionally immortalized murine macrophages (Kiritzy et al., 2021), we show that impairing lipid homeostasis in macrophages at different steps in the process negatively impacts the growth of intracellular *Mtb*, albeit to differing degrees. The impact on *Mtb* growth in the mutant macrophages was mediated through different mechanisms despite some common anti-microbial effectors. *Mtb* infected macrophages deficient in the import of long chain fatty acids increased production of pro-inflammatory markers such as interleukin 1 β (IL-1 β). In contrast, ablation of

lipid droplet biogenesis and fatty acid oxidation increased production of reactive oxygen species and limited the bacteria's access to nutrients. We also found that suppression of *Mtb* growth in macrophages that are unable to produce lipid droplets could not be rescued by exogenous addition of fatty acids indicating that this is not purely nutritional restriction. Our data indicate that interference of lipid metabolism in macrophages leads to suppression of *Mtb* growth via multiple routes.

Results

Knockout of fatty acid import and metabolism genes restricts *Mtb* growth in macrophages

To apply a holistic approach to assessing the role(s) of fatty acid metabolism on the intracellular growth of *Mtb*, we used a CRISPR genetic approach to knockout genes involved in lipid import (CD36, FATP1), lipid droplet formation (PLIN2) and fatty acid oxidation (CPT1A, CPT2) in mouse primary macrophages (Figure 1A). Deletion of CD36 or CPT2 from mouse macrophages has been shown to impair intracellular growth of *Mtb* (Hawkes et al., 2010; Chandra et al., 2020). But the role of specialized long chain fatty acid transporters (FATP1-6) on *Mtb* growth in macrophages is uncharacterized. FATP1 and FATP4 are the most abundant fatty acid transporter isoforms in macrophages (Nishiyama et al., 2018). PLIN2, or adipophilin, is known to be required for lipid droplet formation (Paul et al., 2008; Larigauderie et al., 2004). Five isoforms of mammalian perilipins (PLIN) are involved in lipid droplet biogenesis amongst which PLIN2 is the dominant isoform expressed in macrophages (Knight et al., 2018). However, macrophages derived from PLIN2^{-/-} mice show no defects in production of lipid droplets nor do they impair intracellular growth of *Mtb* (Knight et al., 2018). We targeted each of these genes with at least 2 sgRNAs in Hoxb8 Cas9⁺ conditionally immortalized myeloid progenitors (Kiritsey et al., 2021) to generate a panel of mutants that were deficient in the following candidates of interest; FATP1^{-/-}, PLIN2^{-/-}, CD36^{-/-}, CPT1A^{-/-} and CPT2^{-/-}. Each individual sgRNA achieved >85% CRISPR-mediated deletion efficiency for all the 5 genes, as analyzed by the Inference for CRISPR Edits (ICE) tool (Conant et al., 2022) (Supplementary file 1). We verified the protein knockout phenotypes by flow cytometry and western blot analysis of differentiated macrophages derived from the CRISPR deleted Hoxb8 myeloid precursors (Figure 1 – figure supplement 1).

To confirm certain knockout phenotypes functionally, we checked lipid droplet biogenesis in PLIN2^{-/-} macrophages in comparison to macrophages transduced with a non-targeting scramble sgRNA by confocal microscopy of BODIPY stained cells. Cells were cultured for 24 hours in the presence of exogenous oleate to enhance the formation of lipid droplets (Listenberger and Brown, 2007). We observed a complete absence of lipid droplet formation in PLIN2^{-/-} macrophages as compared to controls (Figure 1 - figure supplement 2A). This is contrary to previous observations in macrophages derived from PLIN2 knockout mice which were reported to have no defect in lipid droplet formation (Knight et al., 2018). We also assessed the ability of CPT2^{-/-} macrophages to oxidize fatty acids using the Agilent Seahorse XF Palmitate Oxidation Stress Test. Scrambled sgRNA and CPT2^{-/-} macrophages were cultured in substrate limiting conditions and supplied with either bovine serum albumin (BSA) or BSA conjugated palmitate. As shown in Figure 1 - figure supplement 2B, control macrophages were able to utilize and oxidize palmitate in substrate limiting conditions indicated by a significant increase in oxygen consumption rates (OCRs) in contrast to cells supplied with BSA alone. Addition of the CPT1A inhibitor, etomoxir, inhibited the cell's ability to use palmitate in these conditions (Figure 1 - figure supplement 2B). CPT2 knockout in CPT2^{-/-} macrophages impaired the cell's ability to oxidize palmitate to a degree comparable to etomoxir treatment as evidenced by baseline OCRs when compared to scrambled sgRNA control (Figure 1 - figure supplement 2C).

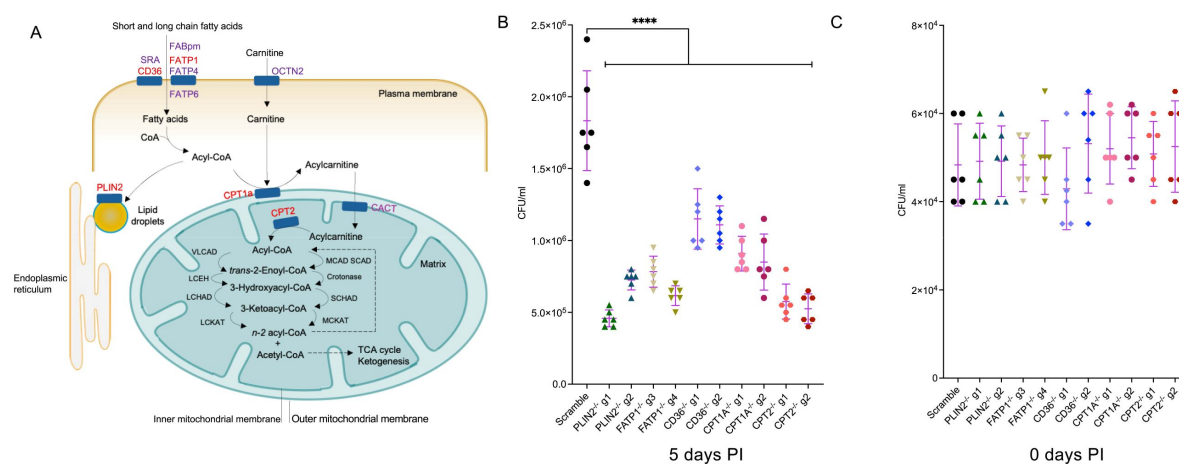


Figure 1.

Knockout of fatty acid import and metabolism genes restricts *Mtb* growth in macrophages.

(A) Schematic of lipid import and metabolism genes in macrophages. Genes targeted for CRISPR-Cas9 mediated knockout are highlighted in red. (B) Scramble sgRNA or specific mutant macrophages were infected with the *Mtb* Erdman strain at MOI 0.4. Intracellular *Mtb* growth was measured by plating and counting CFUs in lysed macrophages 5 days post infection (PI). (C) CFUs from lysed macrophages were also plated on day 0, 3 hours post infection to demonstrate that there were minimal differences in bacterial uptake between the mutated macrophage lineages. n=6; ****, $P < 0.0001$.

We then assessed the different knockout mutant macrophages in their ability to support the intracellular growth of *Mtb*. We infected macrophages with *Mtb* Erdman at a multiplicity of infection (MOI) of 0.4 and assessed intracellular bacterial growth rates by counting colony forming units (CFUs) 5 days post infection. All the 5 mutant macrophages significantly impaired *Mtb* growth rates when compared to scrambled sgRNA as assessed by CFUs counts on day 5 (**Figure 1B**). *PLIN2*^{-/-}, *FATP1*^{-/-} and *CPT2*^{-/-} macrophages displayed the strongest growth restriction phenotypes while *CD36*^{-/-} macrophages had a moderate, but significant, impact on *Mtb* growth. In parallel, we quantified intracellular bacteria on day 0, 3 hours post infection (**Figure 1C**), to ascertain that subsequent differences on day 5 were not due to any disparity in initial bacterial uptake. The moderate growth restriction phenotypes of *CD36*^{-/-} macrophages were consistent with previous findings which reported similar impact on *Mtb* and *M. marinum* growth in macrophages derived from *CD36*^{-/-} mice (Hawkes et al., 2010). Impaired growth of *Mtb* in *CPT1A*^{-/-} and *CPT2*^{-/-} macrophages is also consistent with previous reports that genetic and chemical inhibition of fatty acid oxidation is detrimental to the growth of *Mtb* within macrophages (Huang et al., 2018; Chandra et al., 2020).

***Mtb* infected macrophages with impaired fatty acid import and metabolism display altered mitochondrial metabolism and elevated glycolysis**

Impairment of fatty acids metabolism by *FATP1* knockout in macrophages rewires their substrate bias from fatty acids to glucose (Johnson et al., 2016). We reasoned that deletion of genes required for downstream processing of lipids (**Figure 1A**) could also reprogram macrophages and increase glycolysis which could, in part, explain bacterial growth restriction. We analyzed the metabolic states of 3 knockout macrophages (*FATP1*^{-/-}, *PLIN2*^{-/-} and *CPT2*^{-/-}) in uninfected and *Mtb* infected conditions by monitoring oxygen consumption rates (OCR) and extracellular acidification rates (ECARs) using the Agilent Mito and Glucose Stress Test kits. All 3 mutant uninfected macrophages displayed reduced mitochondrial respiration as evidenced by lower basal and spare respiratory capacity (SRC) when compared to scrambled sgRNA controls (Figure 2 - figure supplement 1A, 1B). *Mtb* infection proportionally reduced basal and SRC rates across all the mutant macrophages and scrambled controls (**Figure 2A, 2B**) when compared to uninfected macrophages, which is consistent with previous findings (Cumming et al., 2018). *PLIN2*^{-/-} macrophages displayed the most marked reduction in mitochondrial activity in both uninfected and infected conditions, while *FATP1*^{-/-} macrophages were the least affected (**Figure 2A**, Figure 2 - figure supplement 1A). As reported previously (Johnson et al., 2016), uninfected *FATP1*^{-/-} macrophages were more glycolytically active with higher basal and spare glycolytic capacity (SGC) compared to scrambled controls (Figure 2 - figure supplement 1C, 1D). Uninfected *PLIN2*^{-/-} and *CPT2*^{-/-} macrophages were also more glycolytically active, but to a greater degree than *FATP1*^{-/-} (Figure 2 - figure supplement 1C, 1D). *Mtb* infection increased the glycolytic capacity of all the 3 mutant macrophages (**Figure 2C, 2D**). Overall, *PLIN2*^{-/-} macrophages exhibited the highest glycolytic capacity (**Figure 2C**, Figure 2 - figure supplement 1D). Our data indicates that impairment of fatty acid metabolism at different steps significantly impacts mitochondrial respiration and reprograms cells towards glycolysis. Increased glycolytic flux in macrophages has been linked to the control of intracellular *Mtb* growth (Gleeson et al., 2016; Shi et al., 2015). Metabolic realignment as a consequence of interference with lipid homeostasis, which results in enhanced glycolysis may contribute to *Mtb* growth restriction in these mutant macrophages.

Knockout of lipid import and metabolism genes in macrophages activates AMPK and stabilizes HIF-1α

Mtb infection is known to induce increased glycolysis or the “Warburg effect” in macrophages, mouse lungs and human TB granulomas (Shi et al., 2015; Gleeson et al., 2016; Belton et al., 2016). Several studies have demonstrated that the Warburg effect is mediated by the master

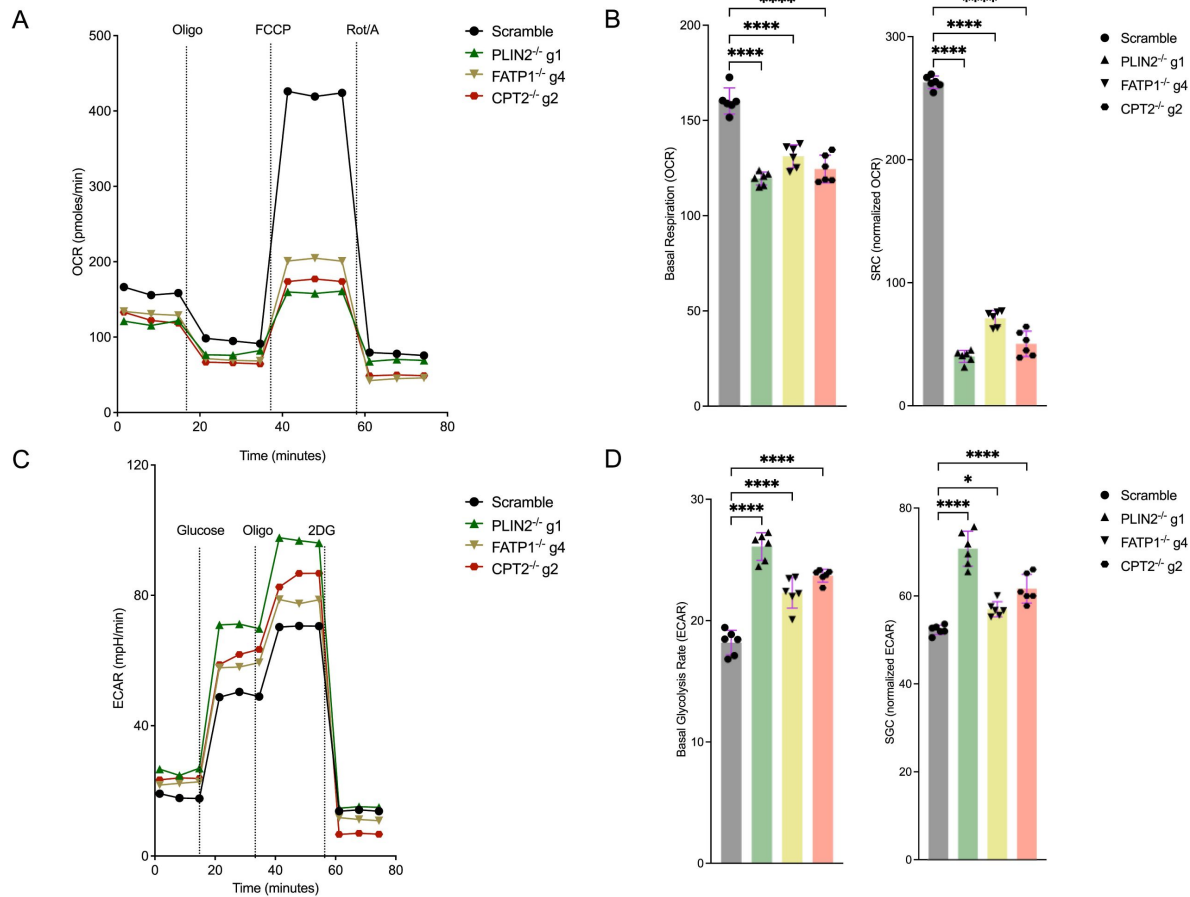


Figure 2.

***Mtb* infected macrophages with impaired fatty acid import and metabolism display reduced mitochondrial activities and are more glycolytic.**

(A) Seahorse flux analyses of scramble or PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages infected with *Mtb* Erdman strain at MOI 1 24 hours post infection. Oxygen consumption rates (OCR) were measured using the Cell Mito Stress Test Kit (Agilent). Oligo, oligomycin; FCCP, fluoro-carbonyl cyanide phenylhydrazone; Rot/A, rotenone and antimycin A. (B) Comparison of basal respiration and spare respiratory capacity (SRC) from A. SRC was calculated by subtracting the normalized maximal OCR from basal OCR. n=3 (2 technical repeats per replicate); ****, $P < 0.0001$. (C) Extracellular acidification rates (ECARs) of scramble or PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages infected with *Mtb* as in A. ECARs were measured using the Agilent Seahorse Glycolysis Stress Test kit. 2DG, 2-Deoxy-D-glucose. (D) Comparison of basal glycolysis and spare glycolytic capacity (SGC) in the indicated mutant macrophages. SGC was calculated as SRC above. n=3 (2 technical repeats per replicate); *, $P < 0.05$, ****, $P < 0.0001$.

transcription factor hypoxia-inducible factor-1 (HIF-1) (Courtney et al., 2015 [DOI](#)). During *Mtb* infection, HIF-1 is activated by production of reactive oxygen species (ROS), TCA cycle intermediates and hypoxia in the cellular microenvironments as a consequence of altered metabolic activities and increased immune cell functions (Li et al., 2024 [DOI](#); Shi et al., 2015 [DOI](#); Gleeson et al., 2016 [DOI](#); Belton et al., 2016 [DOI](#)). We assessed HIF-1 stability in the 3 mutant macrophage lineages (FATP1^{-/-}, PLIN2^{-/-} and CPT2^{-/-}) by monitoring total HIF-1α protein levels by western blot, having confirmed that they were all more glycolytically active than the scrambled controls (Figure 2 [DOI](#), Figure 2 - figure supplement 1). Indeed, all the 3 mutant macrophages displayed significantly higher amounts of total HIF-1α when compared to scrambled controls in uninfected conditions and after *Mtb* infection for 48 hours (Figure 3A, 3B [DOI](#)). We also checked the phosphorylation status of the adenosine monophosphate kinase (AMPK), a master regulator of cell energy homeostasis (Garcia and Shaw, 2017 [DOI](#)), in the mutant macrophages since Seahorse flux analyses indicated that they had impaired mitochondrial activities (Figure 2 [DOI](#), Figure 2 - figure supplement 1). Western blot analysis revealed that in both *Mtb* infected and uninfected conditions, impaired fatty acid metabolism correlated with moderate activation of AMPK as indicated by increased AMPK phosphorylation (Figure 3C, 3D [DOI](#)). These data point to a metabolic reprogramming of cells through activation of HIF-1α and AMPK to promote glycolysis. In energetically stressed cellular environments, activated AMPK promotes catabolic processes such as autophagy to maintain nutrient supply and energy homeostasis (Garcia and Shaw, 2017 [DOI](#)). Autophagy is also an innate immune defense mechanism against intracellular *Mtb* in macrophages (Gutierrez et al., 2004 [DOI](#)). We examined the levels of autophagic flux in the mutant macrophages by monitoring LC3I to LC3II conversion by western blot and by qPCR analysis of selected autophagy genes (*AMBRA1*, *ATG7*, *MAP1LC3B* and *ULK1*). We observed a moderate increase in autophagic flux by western blot in uninfected FATP1^{-/-} and CPT2^{-/-} macrophages, which was amplified upon infection with *Mtb* (Figure 2 - figure supplement 2A, 2B). Our qPCR analysis revealed that the 4 autophagy genes were upregulated in both *Mtb* infected and uninfected conditions in all the 3 mutants (Figure 2 - figure supplement 2C, 2D). These data suggest that impaired fatty acid import and metabolism in macrophages could be restricting *Mtb* growth by promoting autophagy. These data agree with previous observations that inhibition of fatty acid oxidation enhances macrophage xenophagic activity which leads to improved control of *Mtb* (Chandra et al., 2020 [DOI](#)).

Exogenous oleate fails to rescue the *Mtb icl1*-deficient mutant in FATP1^{-/-}, PLIN2^{-/-} and CPT2^{-/-} macrophages

The mycobacterial isocitrate lyase (*icl1*) acts as an isocitrate lyase in the glyoxylate shunt and as a methyl-isocitrate lyase in the methyl-citrate cycle (MCC) (Gould et al., 2006 [DOI](#); McKinney et al., 2000 [DOI](#)). *Mtb* uses the MCC to convert propionyl CoA originating from the breakdown of cholesterol rings and β-oxidation of odd chain fatty acids into succinate and pyruvate, which are eventually assimilated into the TCA cycle (Muñoz-Elías et al., 2006 [DOI](#); Griffin et al., 2012 [DOI](#)). The buildup of propionyl CoA is toxic to *Mtb* and the bacteria relies on the MCC together with the incorporation of propionyl CoA to methyl-branched lipids in the cell wall as an internal detoxification system (Muñoz-Elías et al., 2006 [DOI](#); Savvi et al., 2008 [DOI](#)). *Mtb* propionyl CoA toxicity is, in part, due to a cellular imbalance between propionyl CoA and acetyl CoA as an accumulation of the former or paucity of the latter results in the propionyl CoA mediated inhibition of pyruvate dehydrogenase (Lee et al., 2013 [DOI](#)). Consequently, *Mtb icl1* deficient mutants (*Mtb Δicl1*) are unable to grow in media supplemented with cholesterol or propionate, or intracellularly in macrophages (Lee et al., 2013 [DOI](#)). However, this growth inhibition could be rescued both in culture and in macrophages by exogenous supply of acetate or even chain fatty acids which can be oxidized to acetyl-CoA (Lee et al., 2013 [DOI](#)). We took advantage of this metabolic knowledge to assess whether exogenous addition of the even chain fatty acid oleate can rescue the intracellular growth of *Mtb Δicl1* mutants in our CRISPR knockout macrophages. Scrambled controls, FATP1^{-/-}, PLIN2^{-/-} and CPT2^{-/-} macrophages in normal macrophage media or media supplemented with oleate were

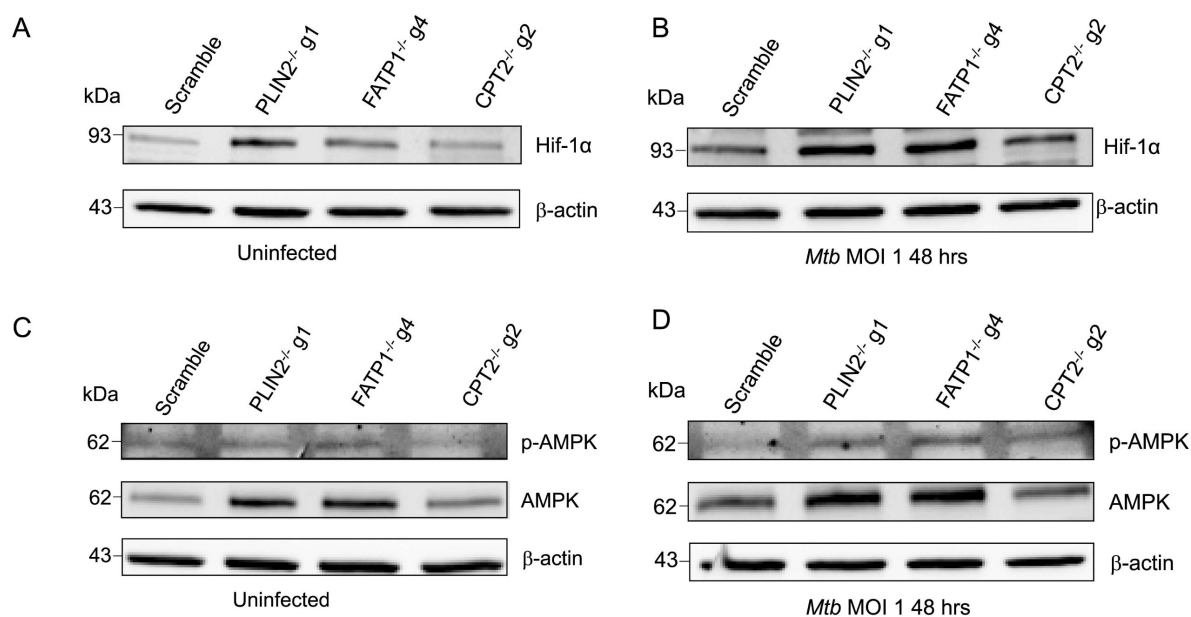


Figure 3.

Knockout of fatty acid import and metabolism genes in macrophages activate AMPK and stabilizes HIF-1α.

(A) Western blot analysis of HIF-1α in uninfected and *Mtb* infected scramble or PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages. In *Mtb* infected conditions, cells were infected with the bacteria at MOI 1 for 48 hours before preparation of cell lysates. (B) Western blot analysis of total and phosphorylated AMPK in uninfected and *Mtb* infected scramble or PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages. Cell lysates were prepared as in A.

infected with an *Mtb Δicl1* strain expressing mCherry at MOI 5. Bacterial growth measured by mCherry expression was recorded 5 days post infection. Consistent with previous observations (Lee et al., 2013 [\[1\]](#)), the *Mtb Δicl1* mutant failed to replicate in both mutant and scramble macrophages that were grown in normal macrophage media as evidenced by baseline mCherry fluorescence (**Figure 4A** [\[1\]](#)). Oleate supplementation successfully rescued the *Mtb Δicl1* mutant in scrambled control macrophages. However, the growth restriction of the *Mtb Δicl1* strain could not be alleviated by exogenous addition of oleate to the mutant macrophages (FATP1^{-/-}, PLIN2^{-/-} and CPT2^{-/-}) (**Figure 4A** [\[1\]](#)). These data suggest that impaired import (FATP1^{-/-}), sequestration (PLIN2^{-/-}) or β-oxidation of fatty acids (CPT2^{-/-}) blocks *Mtb*'s ability to access and use cellular lipids.

Oleate supplementation in macrophages induces the formation of lipid droplets and we were able to confirm the inability to produce lipid droplets in PLIN2^{-/-} macrophages using this approach (Figure 1 - figure supplement 2A). As an indirect measure to track the fate of supplemented oleate in the mutant macrophages, we monitored lipid droplet biogenesis in FATP1^{-/-} and CPT2^{-/-} macrophages to check if the inability to rescue the *Mtb Δicl1* impaired growth phenotypes in these mutant macrophages could be possibly related to disruptions in lipid droplet biogenesis. Confocal analysis of BODIPY stained cells upon oleate supplementation revealed that FATP1^{-/-} macrophages also fail to generate lipid droplets (**Figure 4B** [\[1\]](#)). In contrast, CPT2^{-/-} macrophages produced more and larger lipid droplets in comparison to scrambled controls (**Figure 4B** [\[1\]](#)). These data suggest that the inhibition of *Mtb* growth in these mutant macrophages is not merely through limitation of access to fatty acid nutrients.

Dual RNA sequencing to identify host and bacterial determinants of *Mtb* restriction in mutant macrophage lineages

We performed RNA sequencing of both host and bacteria in *Mtb* infected mutant macrophages as a preliminary step in identification of pathways restricting bacterial growth. We infected scrambled controls, FATP1^{-/-}, PLIN2^{-/-} and CPT2^{-/-} macrophages with the *Mtb smyc*::mCherry strain for 4 days and processed the samples for dual RNA sequencing as previously described (Simwela et al., 2024 [\[2\]](#)). Principal component analysis (PCA) of host transcriptomes revealed a clustering of all the 3 mutant macrophages away from scrambled controls (**Figure 5A** [\[1\]](#)). Interestingly, there was a separation in transcriptional responses between the 3 mutant macrophages as CPT2^{-/-} and PLIN2^{-/-} macrophages clustered closer together and more distant from FATP1^{-/-} (**Figure 5A** [\[1\]](#)). Overall, using an adjusted p-value < 0.05 and absolute log2 fold change > 1.2, we identified 900 genes which were differentially expressed (DE) in PLIN2^{-/-} macrophages (589 up, 311 down), 817 genes which were DE in FATP1^{-/-} macrophages (501 up, 315 down) and 189 genes which were DE in CPT2^{-/-} (124 up, 65 down) (Supplementary file 2, **Figure 5B** [\[1\]](#)). Consistent with the PCA analysis, Venn diagram of the DE genes (**Figure 5B** [\[1\]](#)) indicated divergent responses in the 3 mutant macrophage populations. We performed pathway enrichment analysis (Wu et al., 2021 [\[3\]](#)) of the DE genes to identify anti-microbial pathway candidates in the 3 mutant macrophages. We found that defects in fatty acid uptake in FATP1^{-/-} infected macrophages upregulated pro-inflammatory pathways involved in MAPK and ERK signaling and production of inflammatory cytokines (IFN-γ, interleukin-6, Interleukin-1α, β) (**Figure 5D** [\[1\]](#), Supplementary file 3). The pro-inflammatory signatures of the FATP1^{-/-} macrophages are consistent with previous observations that demonstrated that a deficiency in FATP1 exacerbated macrophage activation *in vitro* and *in vivo* (Johnson et al., 2016 [\[4\]](#)). FATP1 is a solute carrier family member (SLC27A1) and *Mtb* infected FATP1^{-/-} macrophages showed reduced expression of other solute carrier transporters (SLC) such as FATP4 (SLC27A4), GLUT1 (SLC2A1) and 8 SLC amino acid transporters (Figure 5 - figure supplement 1A, 1B, 2A, Supplementary file 2). Interestingly, *Mtb* infected FATP1^{-/-} macrophage transcriptomes exhibited upregulation of macrophage scavenger receptors (MSR1) and the ATP binding cassette transporter ABCC1 (Figure 5 - figure supplement 1A), both of which can independently transport fatty acids into cells (Vogel et al., 2022 [\[5\]](#); Raggars et al., 1999 [\[6\]](#)).

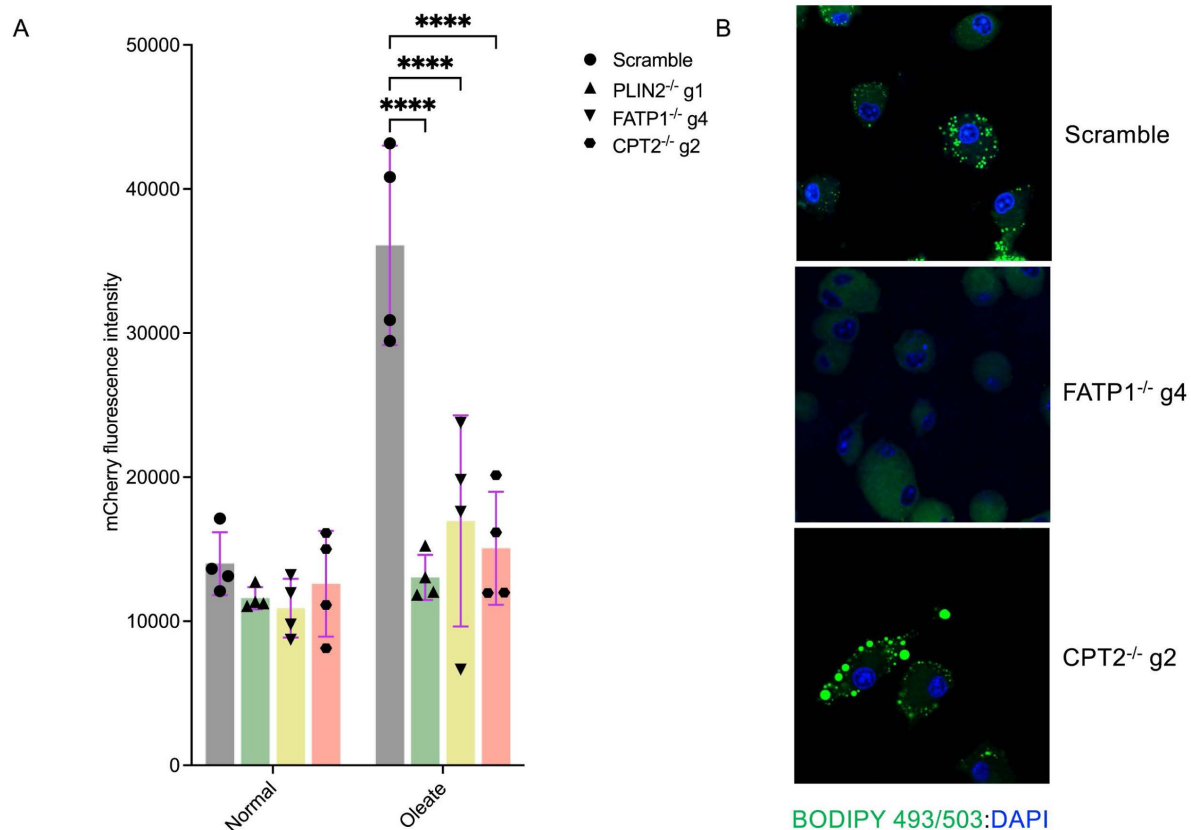


Figure 4.

Supplementation with exogenous oleate fails to rescue the *Mtb* Δ icl1 mutant in PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages.

(A) Scramble or indicated mutant macrophages were infected with the *Mtb* H37Rv Δ icl1 mutant expressing mCherry at MOI 5. Oleate supplementation (400 μ M) was commenced 24 hours before infection in the treatment group, removed during *Mtb* infection and re-added 3-hours post infection for the entire duration of the experiment. Growth kinetics of *Mtb* were measured by monitoring mCherry fluorescence using a plate reader. n=4; ****, $P < 0.0001$. (B) Uninfected scramble or FATP1^{-/-} and CPT2^{-/-} macrophages were supplemented with 400 μ M oleate for 24 hours. Cells were then fixed for 20 minutes and stained for lipid droplet inclusions using the Bodipy 493/503 dye. DAPI was used as a counterstain to detect nuclei.

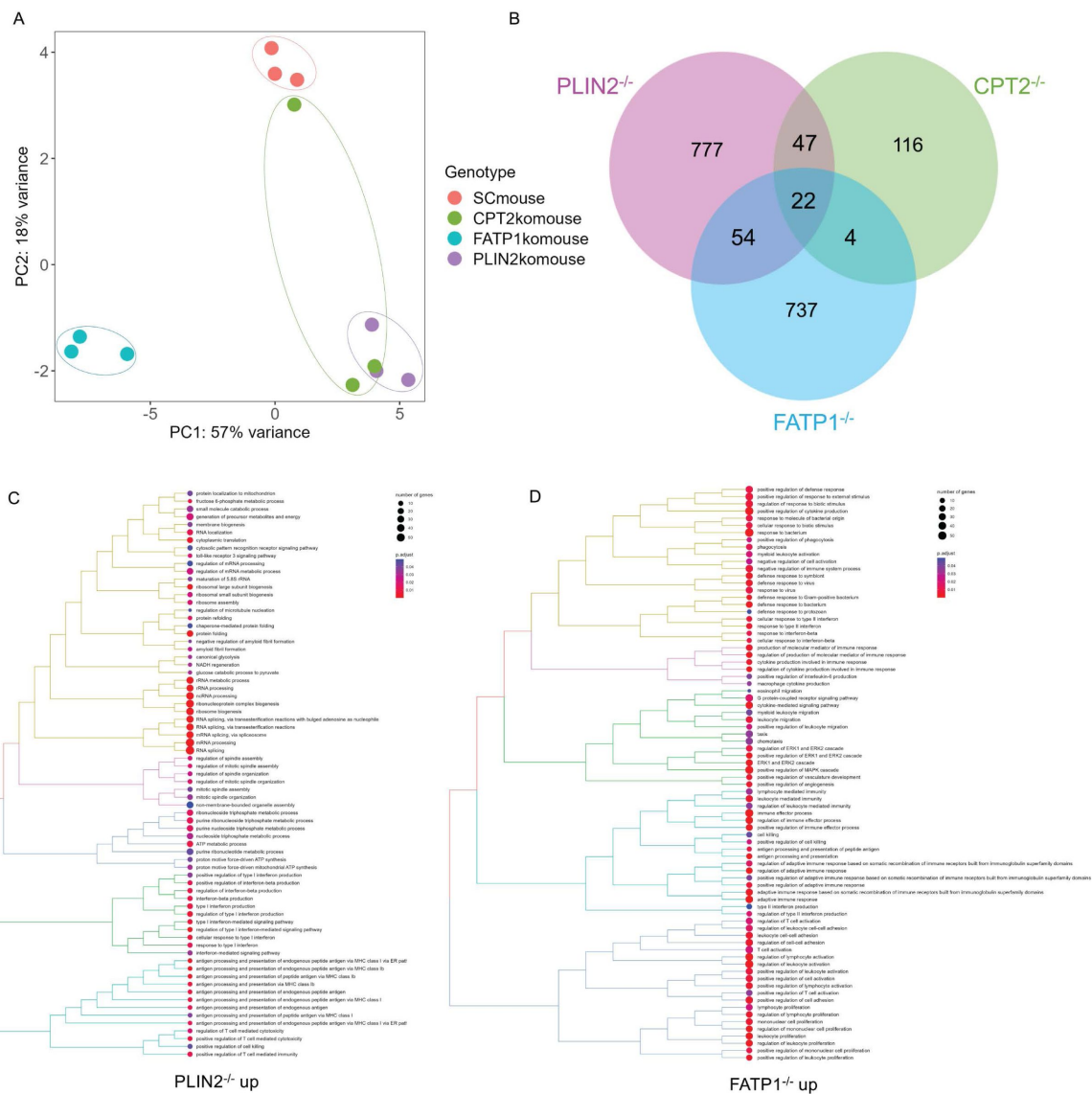


Figure 5.

Dual RNA sequencing to identify host and bacterial determinants of *Mtb* restriction in macrophages with fatty acid import and metabolism knockout genes.

(A) Principal component analysis (PCA) of scramble or PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages transcriptomes infected with the *Mtb* smyc⁺::mCherry strain at MOI 0.5 4 days post infection. (B) Venn diagram of DE gene sets (Supplementary file 2) in PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} mutant macrophages as compared to scramble showing overlapping genes. DE genes cutoff; abs (log₂ fold change) > 0.3, adjusted p-value < 0.05. (C-D) Tree plot of top 80 enriched gene ontology terms (biological process) in *Mtb* infected PLIN2^{-/-} and FATP1^{-/-} upregulated genes.

Meanwhile, inability generate lipid droplets in *Mtb* infected PLIN2^{-/-} macrophages led to upregulation in pathways involved in ribosomal biology, MHC class 1 antigen presentation, canonical glycolysis, ATP metabolic processes and type 1 interferon responses (**Figure 5C** [↗](#), Supplementary file 3). In the downregulated PLIN2^{-/-} DE gene set, enriched pathways included those involved in production of pro-inflammatory cytokines; interleukin-6 and 8, IFN-γ and interleukin-1 (Figure 5 - figure supplement 2B, Supplementary file 3). Oxidative phosphorylation and processes involved in the respiratory chain electron transport were also significantly enriched in PLIN2^{-/-} downregulated genes. This suggests that *Mtb* infected PLIN2^{-/-} macrophages increase glycolytic flux and decrease mitochondrial activities, which is consistent with our metabolic flux analysis data (**Figure 2** [↗](#)). Unlike FATP1^{-/-} macrophages, PLIN2^{-/-} macrophages are, however, broadly anti-inflammatory as most pro-inflammatory genes were downregulated upon *Mtb* infection.

Further downstream in the lipid processing steps, inhibition of fatty acid oxidation in CPT2^{-/-} macrophages upregulated pathways involved in MHC class 1 antigen presentation, response to IFN-γ and interleukin-1 and T-cell mediated immunity (Figure 5 - figure supplement 3, Supplementary file 3). There was a limited overlap in enriched pathways in the upregulated genes between *Mtb* infected CPT2^{-/-} and FATP1^{-/-} macrophages such as those involved in the cellular responses to interleukin-1 and IFN-γ. However, many pathways over-represented in CPT2^{-/-} macrophages were common to PLIN2^{-/-} macrophages (**Figure 5C** [↗](#), Figure 5 - figure supplement 3, Supplementary file 3). Similarly, both *Mtb*-infected PLIN2^{-/-} and CPT2^{-/-} macrophages were downregulated in expression of genes involved in oxidative phosphorylation (Supplementary file 3). We confirmed expression levels of key genes by qPCR analysis of IL-1β and the type 1 interferon (IFN-β) response during the early time points of infection. Indeed, 4 hours post infection, IL-1β and IFN-β were both upregulated in FATP1^{-/-} macrophages as compared to scrambled controls consistent with their pro-inflammatory phenotype (Figure 5 - figure supplement 4A, 4B). On the contrary, PLIN2^{-/-} macrophages downregulated IL-1β (Figure 5 - figure supplement 4B). These data indicate that macrophages respond quite divergently to the deletion of the different steps in fatty acid uptake, which implies that the intracellular pressures to which *Mtb* is exposed may also differ.

Oxidative stress and nutrient limitation are major stresses experienced by *Mtb* in PLIN2^{-/-} and CPT2^{-/-} macrophages

We also analyzed transcriptomes from intracellular *Mtb* from scrambled controls, FATP1^{-/-}, CPT2^{-/-} and PLIN2^{-/-} macrophages in parallel with host transcriptomes in **Figure 5A** [↗](#). Using an adjusted p-value of < 0.1 and an absolute log2 fold change > 1.4, 0 genes were DE in FATP1^{-/-} macrophages, 105 *Mtb* genes were DE in PLIN2^{-/-} macrophages (69 up, 36 down) and 10 genes were DE in CPT2^{-/-} macrophages (3 up, 7 down) (**Figure 6A** [↗](#)). Despite being restrictive to *Mtb* growth (**Figure 1B** [↗](#)) and appearing more pro-inflammatory (**Figure 5D** [↗](#)), FATP1^{-/-} macrophages did not elicit a detectable shift in the transcriptional response in *Mtb*, compared to control host cells. We speculate that pro-inflammatory responses in FATP1^{-/-} macrophages could be enough to restrict the growth of bacteria, but the resulting compensatory responses as evidenced by the upregulation of macrophage scavenger receptors (Figure 5 - figure supplement 1A) alleviate some of the stresses that a lack of fatty acid import could be duly exerting on the bacteria. PLIN2^{-/-} macrophages appeared to elicit the strongest transcriptional response from *Mtb* which is consistent with our CFU data (**Figure 1B** [↗](#)) as PLIN2^{-/-} macrophages exhibited the strongest growth restriction. Among the DE genes in *Mtb* from PLIN2^{-/-} macrophages, a significant number of upregulated genes are involved in nutrient assimilation (**Figure 6B** [↗](#)). *Mtb* in PLIN2^{-/-} macrophages upregulated CobU (Rv0254c) which is predicted to be involved in the bacteria's cobalamin (Vitamin B₁₂) biosynthesis. Vitamin B₁₂ is an important cofactor for the activity of *Mtb* genes required for cholesterol and fatty acid utilization (Campos-Pardos et al., 2024 [↗](#); Sawi et al., 2008 [↗](#)). Genes involved in de novo long chain fatty acid synthesis (AccE5, Rv281) (Bazet Lyonnet et al., 2014 [↗](#)), cholesterol breakdown (HsaD, Rv3569c) (Lack et al., 2010 [↗](#)), β-oxidation of fatty

acids (EchA18, Rv3373; FadE22, Rv3061c) (Schnappinger et al., 2003 [\[1\]](#)), purine salvage (Apt, Rv2584c) (Warner et al., 2014 [\[2\]](#)) and tryptophan metabolism (Lott, 2020 [\[3\]](#)) were also upregulated in *Mtb* from PLIN2^{-/-} macrophages (Figure 6B [\[4\]](#)). This metabolic re-alignment response is seen most frequently under nutrient limiting conditions (Huang et al., 2018 [\[5\]](#); Pisu et al., 2020 [\[6\]](#); Theriault et al., 2022 [\[7\]](#)). *Mtb* in PLIN2^{-/-} macrophages also appears to experience a significant level of other cellular stresses as genes involved in DNA synthesis and repair, general response to oxidative stress and PH survival in the phagosome (DnaN, Rv0002; RecF, Rv0003; DinF, Rv2836c; Rv3242c, Rv1264) were upregulated (Figure 6B [\[4\]](#)). Amongst the down-regulated genes in *Mtb* in PLIN2^{-/-} macrophages, FurA (Rv1909c), a KatG repressor was the most significant (Figure 6B [\[4\]](#), Supplementary file 4). FurA downregulation derepresses the catalase peroxidase, KatG, which promotes *Mtb* survival in oxidative stress conditions (Zahrt et al., 2001 [\[8\]](#)). These data suggest that PLIN2^{-/-} macrophages could be, in part, restricting *Mtb* growth by increasing production of ROS. The data also suggest that, contrary to a previous report (Knight et al., 2018 [\[9\]](#)), blocking lipid droplet formation in host macrophages does place increased nutritional and oxidative stress on intracellular *Mtb*.

CPT2^{-/-} macrophages elicited a modest shift in transcriptional response from *Mtb*, and the majority of the DE genes (8 out of 10) were also DE in PLIN2^{-/-} macrophages (Supplementary file 4). This is in agreement with the host transcription response as CPT2^{-/-} and PLIN2^{-/-} macrophages share similar candidate anti-bacterial responses (Figure 5A-C [\[10\]](#), Figure 5 - figure supplement 3). In common with the *Mtb* transcriptome response in PLIN2^{-/-} macrophages, upregulated genes in *Mtb* isolated from CPT2^{-/-} macrophages included those involved in response to oxidative stress (DinF, Rv2836c) (Supplementary file 4). To substantiate some of these pathways, we assessed the levels of total cellular ROS in FATP1^{-/-}, PLIN2^{-/-} and CPT2^{-/-} macrophages in both *Mtb* infected and uninfected conditions by staining the cells with the Invitrogen CellROX dye and confocal microscopy analysis of live stained cells. In both infected and uninfected conditions, all the 3 mutants displayed significantly elevated total cellular ROS when compared to scramble controls (Fig. 6 - figure supplement 1A, 1B).

Inhibitors of lipid metabolism block intracellular growth of *Mtb* in macrophages but not in broth culture

We next examined if pharmacological inhibitors would phenocopy the growth inhibition phenotypes we observed with specific gene deletions. Compounds which modulate lipid homeostasis are currently being investigated for HDT against TB, which is an area of considerable interest (Kim et al., 2020 [\[11\]](#)). Such compounds include metformin, a widely used anti-diabetic drug which activates AMPK, inhibits fatty acid synthesis and promotes β -oxidation of fatty acids (Fullerton et al., 2013 [\[12\]](#); Singhal et al., 2014 [\[13\]](#)). Chemical inhibition of fatty acid β -oxidation is already known to promote macrophage control of *Mtb* (Chandra et al., 2020 [\[14\]](#); Huang et al., 2018 [\[15\]](#)). We targeted macrophage lipid homeostasis with trimetazidine (TMZ), an inhibitor of β -oxidation of fatty acids, metformin, and an FATP1 inhibitor, FATP1 In (Matsufuji et al., 2013 [\[16\]](#)). We assessed the impact of these compounds on extracellular *Mtb* cultured in broth over 9 days in the presence of the inhibitors (DMSO, TMZ; 500 nM, Metformin; 2 mM, FATP1 In; 10 μ M, Rifampicin; 0.5 μ g/ml). None of the 3 lipid metabolism inhibitors had a measurable effect on *Mtb* growth in liquid culture media when compared to DMSO controls (Figure 7A [\[17\]](#)). Treatment with rifampicin completely blocked bacterial growth under the same conditions (Figure 7A [\[17\]](#)). We next infected scrambled sgRNA control macrophages with *Mtb* at MOI 0.5. Inhibitors were added to infected macrophages 3 hours post infection and bacterial CFUs were enumerated 4 days post treatment. Consistent with previous observations (Chandra et al., 2020 [\[14\]](#); Singhal et al., 2014 [\[13\]](#)), TMZ and metformin significantly reduced bacterial loads in macrophages when compared to DMSO controls (Figure 7B [\[18\]](#)). Similarly, FATP1 In also impacted the intracellular growth of *Mtb* (Figure 7B [\[18\]](#)). The results provide independent data that both genetic and chemical modulation of fatty acid metabolism at different steps in the process negatively impacts intracellular growth of *Mtb*.

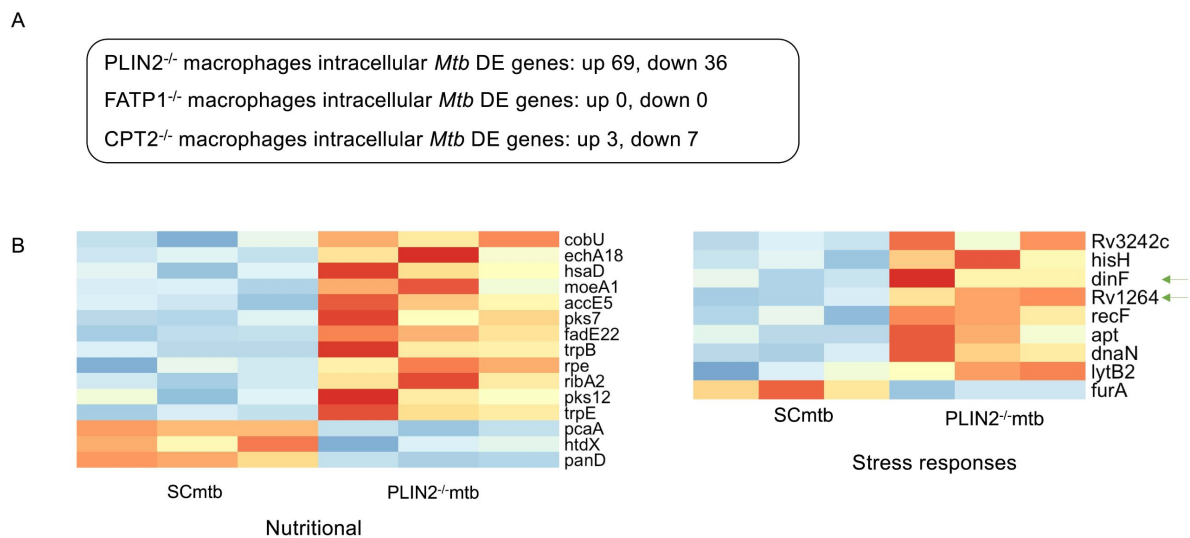


Figure 6.

Nutritional and oxidative stress define the core transcriptome response of *Mtb* inside PLIN2^{-/-} macrophages.

(A) Schematic showing DE genes in *Mtb* transcriptomes isolated from PLIN2^{-/-}, FATP1^{-/-} and CPT2^{-/-} macrophages in **Figure 5A**. (B) Heatmaps of nutritional and oxidative stress DE genes in PLIN2^{-/-} macrophages. Arrows show genes which are also DE in CPT2^{-/-} macrophages in a similar trend (Supplementary file 4).

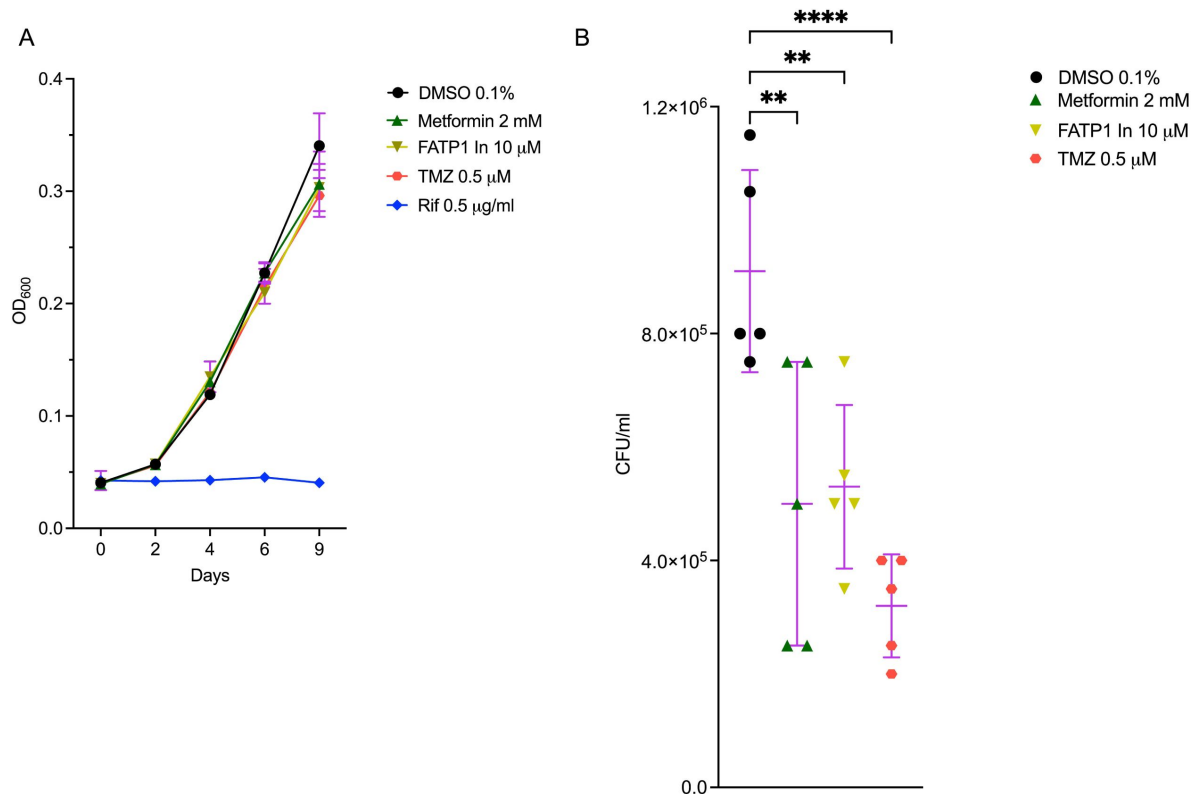


Figure 7.

Inhibitors of fatty acid transport and metabolism block intracellular growth of *Mtb* in macrophages.

(A) Growth of *Mtb* in liquid broth in the absence of drug (DMSO) or presence of metformin, FATP1 inhibitor (FATP1 In, 10 μ M) and the β -oxidation of fatty acid inhibitor, Trimetazidine (TMZ, 500 nM). *Mtb* Erdman was grown to log phase and diluted to OD₆₀₀ 0.01 in 7H9 media in the presence of the above inhibitors. Growth kinetics were monitored by OD₆₀₀ measurements using a plate reader. Rifampicin (RIF) at 0.5 μ g/ml was used as a total killing control. (B) Scramble macrophages were infected with *Mtb* Erdman at MOI 0.5. Inhibitors were added 3 hours post infection following which CFUs were plated 4 days post infection. n=5; **, P < 0.01; ****, P < 0.0001.

Discussion

It is clearly established that host derived fatty acids and cholesterol are important carbon sources for *Mtb* (Wilburn et al., 2018 [DOI](#)). However, the relationship between *Mtb* and the infected host cell lipid metabolism remains a subject of conjecture. In this report, we characterized the role of macrophage lipid metabolism on the intracellular growth of *Mtb* by a targeted CRISPR mediated knockout of host genes involved in fatty acid import, sequestration and catabolism. Macrophage fatty acid uptake is mediated by the scavenger receptor CD36 and specialized long chain fatty acid transporters, FATP1 and FATP4 (Deng et al., 2023 [DOI](#)). Earlier studies reported that a deficiency of CD36 enhances macrophage control of *Mtb*, albeit to a modest degree (Hawkes et al., 2010 [DOI](#)). Work from Hawkes et al. (Hawkes et al., 2010 [DOI](#)) indicated that the anti-microbial effectors in CD36 deficient macrophages were not due to bacterial uptake deficiencies, differences in the rate of *Mtb* induced host cell death, production of ROS or pro-inflammatory cytokines (Hawkes et al., 2010 [DOI](#)). We similarly observed a moderate *Mtb* growth restriction phenotype in our CRISPR generated CD36^{-/-} macrophages. However, a strong growth restriction of *Mtb* was observed when we knocked out the long chain fatty acid transporter, FATP1. FATP1^{-/-} macrophages displayed altered metabolism characterized by stabilization of HIF-1α, activated AMPK, increased glycolysis and reduced mitochondrial functions. Given that both CD36 and FATP1 perform similar functions, it is expected that they should be some degree of compensation between the transporters when either of the genes are deleted. Indeed, we found out that FATP1 knockout resulted in the upregulation of other lipid import transporters (MSR1, ABCC1). This would be consistent with the moderate anti-*Mtb* phenotypes in CD36^{-/-} macrophages which could easily be compensated by the presence of FATPs to alleviate the reduction in fatty acid supply experienced by intracellular *Mtb*. However, the FATP1^{-/-} macrophage phenotype appears to be more severe on *Mtb* and could be exacerbated by an elevated pro-inflammatory response as has been reported previously both *in vitro* and *in vivo* (Johnson et al., 2016 [DOI](#)).

After uptake into the cells, most fatty acids either undergo β-oxidation in the mitochondria to provide energy or are esterified with glycerol phosphate to form triacylglycerols that may be incorporated into lipid droplets in the endoplasmic reticulum (Deng et al., 2023 [DOI](#)). Over the last 2 decades, it has been believed that lipid droplets are a nutrient source for *Mtb* in macrophages (Russell et al., 2009 [DOI](#); Daniel et al., 2011 [DOI](#); Peyron et al., 2008 [DOI](#); Singh et al., 2012 [DOI](#)). However, recent work indicates lipid droplets may serve as centers for the production of pro-inflammatory markers and anti-microbial peptides (Knight et al., 2018 [DOI](#); Bosch et al., 2020 [DOI](#)). It has also been reported that bone marrow macrophages derived from PLIN2^{-/-} mice did not have defects in the formation of lipid droplets and supported robust intracellular *Mtb* replication (Knight et al., 2018 [DOI](#)). However, our mutant macrophages with myeloid specific knockout of PLIN2 are unable to form lipid droplets and are defective in supporting the growth of *Mtb*. The discrepancies with previous observations (Knight et al., 2018 [DOI](#)) could be a consequence of compensatory responses to PLIN2 knockout in whole mice which when performed at embryonic level would allow for sufficient time for the cells to recover by upregulating related PLIN isoforms. In fact, PLIN2^{-/-} macrophages displayed the strongest anti-*Mtb* phenotypes amongst our mutants exhibiting activated AMPK, increased glycolysis and autophagy and impaired mitochondrial functions. *Mtb* isolated from PLIN2^{-/-} macrophages displayed signatures of severe nutrient limitation and oxidative stress damage.

It has also been previously reported that chemical, genetic or miR33 mediated blockade of fatty acid β-oxidation in macrophages induces lipid droplet formation (Chandra et al., 2020 [DOI](#); Quimet et al., 2016 [DOI](#)) but that this enhanced lipid droplet formation does not correlate with improved intracellular *Mtb* growth (Chandra et al., 2020 [DOI](#)). We observed a similar phenotype as CPT2^{-/-} macrophages that generated larger and more abundant lipid droplets than scrambled control

macrophages were still restrictive to *Mtb* growth. This implies that the presence or absence of lipid droplets does not in itself indicate whether a macrophage will support or restrict *Mtb* growth, and that the anti-microbial environment extends beyond simple nutrient availability.

In summary, our study shows that blocking macrophage's ability to import, sequester or catabolize fatty acids chemically or by genetic knockout impairs *Mtb* intracellular growth. There are shared features between potential anti-microbial effectors in macrophages which lack the ability to import (FATP^{-/-}) or metabolize fatty acids (PLIN2^{-/-}, CPT2^{-/-}) such as increased glycolysis, stabilized HIF-1α, activated AMPK, enhanced autophagy and production of ROS. However, there are also intriguing points of divergence as FATP^{-/-} macrophages are more pro-inflammatory while PLIN2^{-/-} macrophages appear to be broadly anti-inflammatory. The routes to *Mtb* growth restriction in these mutant macrophages are clearly more complex than the bacteria's inability to acquire nutrients. The data further emphasizes that targeting fatty acid homeostasis in macrophages at different steps in the process (uptake, storage and catabolism) is worthy of exploring in the development of new therapeutics against tuberculosis.

Materials and methods

All materials and methods are as described (Simwela et al., 2024 [DOI](#)) unless otherwise specified.

Flow cytometry and western blot analysis

Generation of CRISPR mutant Hoxb8 macrophages was carried out described previously (Simwela et al., 2024 [DOI](#)). Antibodies used for both western blot and flow cytometry were as follows: rat anti-Mouse CD36:Alexa Fluor®647 (Biorad, 10 µl/million cells), rabbit anti-PLIN2 (Proteintech, 1:1000), rabbit anti-FATP1 (Affinity biosciences, 1:1000), rabbit anti-CPT1A antibody (Proteintech, 1:1000), rabbit anti-CPT2 antibody (Proteintech, 1:1000), rabbit anti-HIF-1α antibody (Proteintech, 1:1000), rabbit anti-AMPKα(1:1000, Cell signalling Technology), rabbit anti-Phospho-AMPKα(1:500, Cell signalling Technology), rabbit anti-LC3B (1:1000, Cell signalling Technology) and rabbit anti-β Actin (1:1000, Cell Signalling Technology). For western blots, secondary antibodies used were anti-rabbit/mouse StarBright Blue 700 (1:2500, Biorad). Blots were developed and imaged as described previously (Simwela et al., 2024 [DOI](#)).

Staining for cellular lipid droplets

Macrophages monolayers in Ibidi eight-well chambers were supplemented with exogenous 400 µM oleate for 24 hours to induce the formation of lipid droplets (Listenberger and Brown, 2007 [DOI](#)). Cells were then fixed in 4% paraformaldehyde and stained with BODIPY™ 493/503 (Invitrogen, 1 µg/ml) in 150 mM sodium chloride. Stained cells were mounted with media containing DAPI and imaged using a Leica SP5 confocal microscope.

Seahorse XF palmitate oxidation stress test

A modified Seahorse mitochondrial stress test was used to measure macrophage's ability to oxidize palmitate in substrate limiting conditions. 2 days before the assay, 1 x 10⁵ cells were plated in Seahorse cell culture mini plates. 1 day before the assay, macrophage media was replaced with the Seahorse substrate limited growth media (DMEM without pyruvate supplemented with 0.5 mM glucose, 1 mM glutamine, 1% FBS and 0.5 mM L-Carnitine). On the day of the assay, substrate limited media was replaced with assay media (DMEM without pyruvate supplemented with 2 mM glucose and 0.5 mM L-Carnitine). In selected treatment conditions, cells were either supplied with bovine serum albumin (BSA), BSA palmitate or BSA palmitate plus etomoxir (4 µM). Oxygen consumption rates (OCRs) were measured using the Mito Stress Test assay conditions as described previously (Simwela et al., 2024 [DOI](#)).

Rescue of the *Mtb* Δ icl1 mutant in oleate supplemented media

The *Mtb* H37Rv Δ icl1 mutant expressing mCherry (Lee et al., 2013 [DOI](#)) was used for the rescue experiments. The strain was maintained in 7H9 OADC broth as previously described (Simwela et al., 2024 [DOI](#)) in the presence of Kanamycin (25 μ g/ml) and hygromycin (50 μ g/ml). 24 hours before infection, macrophages were cultured in normal macrophage media or media supplemented with 400 μ M oleate. Cells were then infected with the *Mtb* Δ icl1 mutant at MOI 5. The bacterial mCherry signal was measured on day 0 and day 5 post infection on an Envision plate reader (PerkinElmer). Oleate was maintained throughout the experiment in the rescue assay conditions.

Measurement of total cellular ROS

Uninfected or *Mtb* infected macrophages monolayers in Ibidi eight-well chambers were stained with the CellROX Deep Red dye (Invitrogen) as per manufacturer's staining protocol. Live stained cells were imaged on a Leica SP5 confocal microscope. Z-stacks were re-constructed in ImageJ from which mean fluorescence intensities (MFI) for individual cells were obtained.

Datasets

The RNA-seq data from the Dual RNA-seq analysis of infected mouse macrophages,542 which includes both macrophage and *Mtb* reads are available in GEO: GSE270571

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Reviewer #1 (Public review):**Summary:**

This study investigates the role of macrophage lipid metabolism in the intracellular growth of *Mycobacterium tuberculosis*. By using a CRISPR-Cas9 gene-editing approach, the authors knocked out key genes involved in fatty acid import, lipid droplet formation, and fatty acid oxidation in macrophages. Their results show that disrupting various stages of fatty acid metabolism significantly impairs the ability of Mtb to replicate inside macrophages. The mechanisms of growth restriction included increased glycolysis, oxidative stress, pro-inflammatory cytokine production, enhanced autophagy, and nutrient limitation. The study demonstrates that targeting fatty acid homeostasis at different stages of the lipid metabolic process could offer new strategies for host-directed therapies against tuberculosis.

The work is convincing and methodologically strong, combining genetic, metabolic, and transcriptomic analyses to provide deep insights into how host lipid metabolism affects bacterial survival.

Strengths:

The study uses a multifaceted approach, including CRISPR-Cas9 gene knockouts, metabolic assays, and dual RNA sequencing, to assess how various stages of macrophage lipid metabolism affect Mtb growth. The use of CRISPR-Cas9 to selectively knock out key genes involved in fatty acid metabolism enables precise investigation of how each step—lipid import, lipid droplet formation, and fatty acid oxidation—affects Mtb survival. The study offers mechanistic insights into how different impairments in lipid metabolism lead to diverse antimicrobial responses, including glycolysis, oxidative stress, and autophagy. This deepens the understanding of macrophage function in immune defense.

The use of functional assays to validate findings (e.g., metabolic flux analyses, lipid droplet formation assays, and rescue experiments with fatty acid supplementation) strengthens the reliability and applicability of the results.

By highlighting potential targets for HDT that exploit macrophage lipid metabolism to restrict Mtb growth, the work has significant implications for developing new tuberculosis treatments.

Weaknesses:

The experiments were primarily conducted *in vitro* using CRISPR-modified macrophages. While these provide valuable insights, they may not fully replicate the complexity of the *in vivo* environment where multiple cell types and factors influence Mtb infection and immune responses.

<https://doi.org/10.7554/eLife.102980.1.sa3>

Reviewer #2 (Public review):**Summary:**

Host-derived lipids are an important factor during Mtb infection. In this study, using CRISPR knockouts of genes involved in fatty acid uptake and metabolism, the authors claim that a compromised uptake, storage, or metabolism of fatty acid restricts Mtb growth upon infection. Further, the authors claim that the mechanism involves increased glycolysis, autophagy, oxidative stress, pro-inflammatory cytokines, and nutrient limitation. The authors also claim that impaired lipid droplet formation restricts Mtb growth. However, promoting lipid droplet biogenesis does not reverse/promote Mtb growth.

Strengths:

The strength of the study is the use of clean HOXB8-derived primary mouse macrophage lines for generating CRISPR knockouts.

Weaknesses:

There are many weaknesses of this study, they are clubbed into four categories below

- (1) Evidence and interpretations: The results shown in this study at several places do not support the interpretations made or are internally contradictory or inconsistent. There are several important observations, but none were taken forward for in-depth analysis.
 - a) The phenotypes of PLIN2^{-/-}, FATP1^{-/-}, and CPT^{-/-} are comparable in terms of bacterial growth restriction; however, their phenotype in terms of lipid body formation, IL1B expression, etc., are not consistent. These are interesting observations and suggest additional mechanisms specific to specific target genes; however, clubbing them all as altered fatty acid uptake or catabolism-dependent phenotypes takes away this important point.
 - b) Finding the FATP1 transcript in the HOXB8-derived FATP1^{-/-} CRISPR KO line is a bit confusing. There is less than a two-fold decrease in relative transcript abundance in the KO line compared to the WT line, leaving concerns regarding the robustness of other experiments as well using FATP1^{-/-} cells.
 - c) No gene showing differential regulation in FATP^{-/-} macrophages, which is very surprising.
 - d) ROS measurements should be done using flow cytometry and not by microscopy to nail the actual pattern.
- (2) Experimental design: For a few assays, the experimental design is inappropriate
 - a) For autophagy flux assay, immunoblot of LC3II alone is not sufficient to make any interpretation regarding the state of autophagy. This assay must be done with BafA1 or CQ controls to assess the true state of autophagy.
 - b) Similarly, qPCR analyses of autophagy-related gene expression do not reflect anything on the state of autophagy flux.
- (3) Using correlative observations as evidence:
 - a) Observations based on RNAseq analyses are presented as functional readouts, which is incorrect.
 - b) Claiming that the inability to generate lipid droplets in PLIN2^{-/-} cells led to the upregulation of several pathways in the cells is purely correlative, and the causal relationship does not exist in the data presented.
- (4) Novelty: A few main observations described in this study were previously reported. That includes Mtb growth restriction in PLIN2 and FATP1 deficient cells. Similarly, the impact of Metformin and TMZ on intracellular Mtb growth is well-reported. While that validates these observations in this study, it takes away any novelty from the study.
- (5) Manuscript organisation: It will be very helpful to rearrange figures and supplementary figures.

<https://doi.org/10.7554/eLife.102980.1.sa2>

Reviewer #3 (Public review):

Summary:

This study provides significant insights into how host metabolism, specifically lipids, influences the pathogenesis of *Mycobacterium tuberculosis* (Mtb). It builds on existing

knowledge about Mtb's reliance on host lipids and emphasizes the potential of targeting fatty acid metabolism for therapeutic intervention.

Strengths:

To generate the data, the authors use CRISPR technology to precisely disrupt the genes involved in lipid import (CD36, FATP1), lipid droplet formation (PLIN2), and fatty acid oxidation (CPT1A, CPT2) in mouse primary macrophages. The Mtb Erdman strain is used to infect the macrophage mutants. The study, reveals specific roles of different lipid-related genes. Importantly, results challenge previous assumptions about lipid droplet formation and show that macrophage responses to lipid metabolism impairments are complex and multifaceted. The experiments are well-controlled and the data is convincing.

Overall, this well-written paper makes a meaningful contribution to the field of tuberculosis research, particularly in the context of host-directed therapies (HDTs). It suggests that manipulating macrophage metabolism could be an effective strategy to limit Mtb growth.

Weaknesses:

None noted. The manuscript provides important new knowledge that will lead to host-directed therapies to control Mtb infections.

<https://doi.org/10.7554/eLife.102980.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study investigates the role of macrophage lipid metabolism in the intracellular growth of Mycobacterium tuberculosis. By using a CRISPR-Cas9 gene-editing approach, the authors knocked out key genes involved in fatty acid import, lipid droplet formation, and fatty acid oxidation in macrophages. Their results show that disrupting various stages of fatty acid metabolism significantly impairs the ability of Mtb to replicate inside macrophages. The mechanisms of growth restriction included increased glycolysis, oxidative stress, pro-inflammatory cytokine production, enhanced autophagy, and nutrient limitation. The study demonstrates that targeting fatty acid homeostasis at different stages of the lipid metabolic process could offer new strategies for host-directed therapies against tuberculosis.

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and autophagy. This deepens the understanding of macrophage function in immune defense.

The use of functional assays to validate findings (e.g., metabolic flux analyses, lipid droplet formation assays, and rescue experiments with fatty acid supplementation) strengthens the reliability and applicability of the results.

By highlighting potential targets for HDT that exploit macrophage lipid metabolism to restrict Mtb growth, the work has significant implications for developing new tuberculosis treatments.

Weaknesses:

The experiments were primarily conducted in vitro using CRISPR-modified macrophages. While these provide valuable insights, they may not fully replicate the complexity of the in vivo environment where multiple cell types and factors influence Mtb infection and immune responses.

We thank the reviewer for pointing this out. We acknowledge that our in vitro system may indeed not fully replicate the complex in vivo environment in light of the heterogeneous responses of macrophages to Mtb infection in whole animal models. We do believe, however, that the Hoxb8 in vitro model provides a powerful genetic tool to interrogate host-Mtb interactions using primary macrophages that represent the bone marrow-derived macrophage lineage. Reviewer #1 also made several helpful suggestions in their recommendations to authors relating to the reorganization of the data in our Figures in both the manuscript and the supplemental data. We will incorporate these suggestions into the revised version of the manuscript upon resubmission.

Reviewer #2 (Public review):

Summary:

Host-derived lipids are an important factor during Mtb infection. In this study, using CRISPR knockouts of genes involved in fatty acid uptake and metabolism, the authors claim that a compromised uptake, storage, or metabolism of fatty acid restricts Mtb growth upon infection. Further, the authors claim that the mechanism involves increased glycolysis, autophagy, oxidative stress, pro-inflammatory cytokines, and nutrient limitation. The authors also claim that impaired lipid droplet formation restricts Mtb growth. However, promoting lipid droplet biogenesis does not reverse/promote Mtb growth.

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a) The phenotypes of PLIN2^{-/-}, FATP1^{-/-}, and CPT^{-/-} are comparable in terms of bacterial growth restriction; however, their phenotype in terms of lipid body formation, IL1B expression, etc., are not consistent. These are interesting observations and suggest additional mechanisms specific to specific target genes; however, clubbing them all as

altered fatty acid uptake or catabolism-dependent phenotypes takes away this important point.

We thank the reviewer for highlighting this. Our main focus was on assessing the impact of manipulating lipid homeostasis in macrophages and the consequences this has on the intracellular growth of Mtb. It was never our intention to imply these mutants generated equivalent phenotypes, and we will modify the revised manuscript to reflect this point. We will stress that interfering with lipid processing at different stages in macrophages results in both shared and divergent anti-microbial conditions against Mtb.

b) Finding the FATP1 transcript in the HOXB8-derived FATP1^{-/-} CRISPR KO line is a bit confusing. There is less than a two-fold decrease in relative transcript abundance in the KO line compared to the WT line, leaving concerns regarding the robustness of other experiments as well using FATP1^{-/-} cells.

CRISPR-Cas9 targeting of genes with single sgRNAs as is the case with our mutants generates insertions and deletions (INDELs) at the CRISPR cut site. These INDELs do not block mRNA transcription totally, and this is widely reported and accepted in the field. In these cases, RT-PCR or RNA-seq methods are not used to verify CRISPR knockouts as they are not sensitive enough to identify INDELs. We provide knockout efficiencies by ICE analysis in supplemental information file 1 for all the mutants used in the study. We also demonstrate protein depletion by western blot and flow cytometry for all the mutants (Figure 1 - figure supplement 1). Only mutants with greater than >90% protein depletion were used for subsequent characterization.

c) No gene showing differential regulation in FATP1^{-/-} macrophages, which is very surprising.

We assume the reviewer is referring to the Mtb transcriptome response in FATP1^{-/-} macrophages, which we agree was unexpected. However, we saw a significant compensatory response in the host cell (at transcriptional level) in FATP1^{-/-} macrophages as evidenced by an upregulation of other fatty acid transporters (Figure 5 - figure supplement 1). We postulate that these compensatory responses could, in part, alleviate the stresses the bacteria experience within the cell, and these were discussed in the manuscript.

d) ROS measurements should be done using flow cytometry and not by microscopy to nail the actual pattern.

We thank the reviewer for the suggestion. However, confocal imaging is also widely used to measure ROS with similar quantitative power and individual cell resolution (PMID: 32636249, 35737799).

(2) Experimental design: For a few assays, the experimental design is inappropriate

a) For autophagy flux assay, immunoblot of LC3II alone is not sufficient to make any interpretation regarding the state of autophagy. This assay must be done with BafA1 or CQ controls to assess the true state of autophagy.

We would like to point out that monitoring LC3I to LC3II conversion by western blot, confocal imaging of LC3 puncta and qPCR analysis of autophagy related genes are all validated assays for monitoring autophagic flux in a wide variety of cells. We refer the reviewer to the latest extensive guidelines on the subject (PMID: 33634751). Furthermore, Bafilomycin A and chloroquine are not specific inhibitors of autophagy and therefore are of limited value as controls. BafA is an inhibitor of the proton-ATPase apparatus as well impacting autophagy

through activity on the Ca-P60A/SERCA pathway. Chloroquine impacts vacuole acidification, autophagosome/lysosome fusion and slows phagosome maturation. So, while BafA and chloroquine will reduce autophagy their effects are pleiotropic and their impact on Mtb is unknown.

b) Similarly, qPCR analyses of autophagy-related gene expression do not reflect anything on the state of autophagy flux.

See our response above.

(3) Using correlative observations as evidence:

a) Observations based on RNAseq analyses are presented as functional readouts, which is incorrect.

We are not entirely sure where we used our RNA-seq data sets as functional readouts. We used our transcriptome data to provide a preliminary identification of anti-microbial responses in the mutant macrophages infected with Mtb. Where applicable, we followed up and confirmed the more compelling RNA-seq data either by metabolic flux analyzes, qPCR, ROS measurements, and quantitative imaging.

b) Claiming that the inability to generate lipid droplets in PLIN2-/- cells led to the upregulation of several pathways in the cells is purely correlative, and the causal relationship does not exist in the data presented.

Again, it was not our intention to infer causality. Throughout the manuscript, we endeavor to present our data with a specific focus on describing the consequences of interfering with either fatty acid import, lipid droplet biogenesis and fatty acid oxidation on macrophage responses to Mtb. We will revisit the revised manuscript to remove any sections that imply causality.

(4) Novelty: A few main observations described in this study were previously reported. That includes Mtb growth restriction in PLIN2 and FATP1 deficient cells. Similarly, the impact of Metformin and TMZ on intracellular Mtb growth is well-reported. While that validates these observations in this study, it takes away any novelty from the study.

To the best of our knowledge, Mtb growth restrictions in PLIN2 and FATP1 deficient macrophages have not been reported elsewhere. To the contrary, PLIN2 knockout macrophages obtained from PLIN2 deficient mice have been reported to robustly support Mtb replication (PMID: 29370315), quite the opposite to our data. We extensively discuss these discrepancies in the manuscript. We also discuss and cite appropriate references where Mtb growth restriction for similar macrophage mutants have been reported (CD36^{-/-} and CPT2^{-/-}). Our aim was to carry out a systematic myeloid specific genetic interference of fatty acid import, storage and catabolism to assess the effect on Mtb growth at all stages of lipid handling instead of focusing on one target. In the chemical approach, we used TMZ and Metformin deliberately because they had already been reported as being active against intracellular Mtb and we wished to place our data in the context of existing literature. These studies were referenced extensively in the text.

(5) Manuscript organisation: It will be very helpful to rearrange figures and supplementary figures.

We will re-organize the figures in the manuscript revision as per the reviewer's recommendation, and the recommendations of reviewer #1.

We will address the other concerns raised by reviewer #2 in the recommendations to authors during revision of the manuscript.

Reviewer #3 (Public review):

Summary:

This study provides significant insights into how host metabolism, specifically lipids, influences the pathogenesis of Mycobacterium tuberculosis (Mtb). It builds on existing knowledge about Mtb's reliance on host lipids and emphasizes the potential of targeting fatty acid metabolism for therapeutic intervention.

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Weaknesses:

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